

4(a)	to the left/from the right/from B to A/opposite (direction) to (α -particle) velocity	B1
4(b)	$v^2 = u^2 + 2as$ $s = (4.1 \times 10^6)^2 / (2 \times 2.7 \times 10^{14})$	C1
	$= 0.031 \text{ m}$	A1
4(c)	$E = F / Q$ or $E = ma / Q$	C1
	$= (4 \times 1.66 \times 10^{-27} \times 2.7 \times 10^{14}) / (2 \times 1.60 \times 10^{-19})$	C1
	$= 5.6 \times 10^6 \text{ V m}^{-1}$	A1
4(d)	straight line with negative gradient that intercepts both the momentum and t axes	B1
4(e)	force (on α -particle)	B1
4(f)(i)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 9.11 \times 10^{-31} \times (4.1 \times 10^6)^2$	C1
	$= 7.7 \times 10^{-18} \text{ J}$	A1
4(f)(ii)	particles have opposite charges	B1
	(so) forces (on charges) are opposite (directions)	B1
	β^- has less/half the charge so less/half the force	B1
4(f)(iii)	(electron) <u>antineutrino</u>	B1